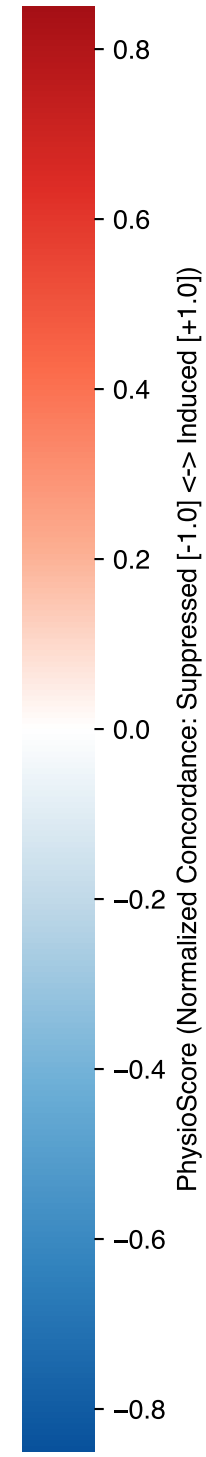
Figure 5: PhysioSpace Stress Pattern Autodecoding across the Suborbital Flight Profile
Dynamic activation of 20 canonical Arabidopsis stress programs (* p<0.05, ** p<0.01, *** p<0.001)

LEAVES (Blue Origin NS-26) | ROOTS (Blue Origin NS-26) | COMPARATIVE

	EE Ascent (~2.5m)	CS Micro-g (~4m)	RE Coast (~7.5m)	LN Touchdown (~10m)	EE Ascent (~2.5m)	CS Micro-g (~4m)	RE Coast (~7.5m)	LN Touchdown (~10m)	Virgin Galactic Unity 22
Heat	+0.55**	+0.68***	+0.77***	+0.61***	-0.04	+0.56***	+0.74***	+1.00*	+0.98***
Hypoxia	+0.01	-0.09	+0.14	+0.10	0.00	+0.47***	+0.48***	+1.00*	-0.04
Drought	+0.59**	+0.56***	+0.45***	+0.43***	+0.03	+0.38**	+0.56***	+1.00*	+0.77***
Osmotic	-0.16	+0.35***	+0.24**	+0.21*	+0.68	+0.42**	+0.52***	+0.89	+0.59***
Salt	-0.02	+0.36***	+0.34***	+0.33***	-0.04	+0.75***	+0.52***	+0.56	+0.54***
Cold	+0.13	-0.10	+0.05	+0.08	-0.11	-0.30*	-0.26	-0.11	+0.09
Drought.Light	-0.07	+0.36***	+0.26**	+0.21*	+0.07	+0.27*	+0.04	+0.89	+0.34*
Wounding	-0.55**	-0.07	+0.04	-0.10	-0.21	+0.03	+0.43**	+0.78	+0.01
Genotoxic	-0.55**	+0.10	-0.17**	-0.16	-0.05	+0.01	-0.39**	-0.22	+0.08
Radiation	-0.51**	+0.17	+0.11	+0.21*	+0.27	-0.02	+0.12	+0.78	+0.15
UV	-0.44*	+0.51***	+0.29***	+0.16	+0.12	+0.24	+0.55***	+0.78	+0.52***
Herbicide	-0.05	-0.07	-0.06	-0.02	+0.09	-0.28*	-0.20	-0.22	-0.03
LowPH	-0.16	+0.22	+0.21*	+0.08	-0.06	+0.51***	+0.11	+0.33	+0.32
Biotic	-0.49**	+0.29**	-0.07	0.00	+0.47	-0.27*	-0.33*	-0.11	-0.04
Biotic.Hormone	+0.10	+0.49***	+0.19*	+0.16	-0.66	-0.03	+0.18	+0.78	+0.40*
Hormone	+0.01	+0.08	+0.20*	+0.17	-0.02	+0.06	+0.10	+0.56	+0.33
Light	+0.46*	+0.69***	+0.53***	+0.34***	-0.13	+0.63***	+0.66***	+1.00*	+0.77***
FarRed	+0.41*	-0.09	+0.27***	+0.38***	-0.56	+0.08	+0.29*	+1.00*	+0.19
Nitrogen	+0.26	-0.18**	-0.06	+0.03	-0.03	-0.21	+0.18	+0.44	-0.35***
FeDeficiency	-0.56**	+0.43***	+0.10	+0.04	+0.08	+0.06	+0.02	0.00	+0.17



Color scale: (Low) Deep Blue = Suppressed Program <-> White = Baseline <-> (High) Crimson Red = Induced Program.
Projections performed against canonical AT_Stress_Space (20 axes x 22,729 genes; Hadizadeh Esfahani et al. 2021).